

Revisiting the Etiology of Cocaine-Related Ischemic Strokes: An Observational Cohort

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Dear Sir:

Chronic cocaine use increases the odds of ischemic stroke, intracerebral hemorrhage, and subarachnoid hemorrhage, while additionally increasing overall mortality risk.¹ The proposed mechanism of vasospasm causing small vessel occlusion in the brain has been credited as the culprit behind acute ischemic strokes (AIS) in patients who use cocaine.² Few studies have confirmed these unique stroke mechanisms associated with acute and chronic cocaine use, and further validation is required.³ Additionally, the safety of using combination thrombolytic therapy and endovascular thrombectomy (EVT) in these patients has not been well-studied.⁴

Understanding the unique causes of cerebral infarction associated with cocaine use may inform future treatment decisions and provide evidence supporting the use of acute reperfusion therapies, such as EVT, in this population.

This retrospective observational cohort study was conducted at a comprehensive stroke center between January 2021 and May 2022. Approval for this study was granted through the institutional review board (IRB#21-1903) with a waiver of informed consent.

Patients were required to have a primary diagnosis of AIS and a positive urine toxicology test for cocaine. Patient demographics, substance use history, urine toxicology upon admission, stroke type (hemorrhagic vs. ischemic), TOAST (Trial of ORG 10172 in Acute Stroke Treatment) criteria classification,⁵ past medical history, current medications, initial lab work, initial and discharge National Institutes of Health Stroke Scale (NIHSS) scores, baseline modified Rankin Scale (mRS) score at discharge or earliest follow-up, imaging results, interventions, and overall mortality

were abstracted. Descriptive statistics (i.e., mean, median, standard deviation) were conducted using IBM SPSS Version 28 (IBM Corp., Armonk, NY, USA).

Seventy-four patients were identified for analysis. Baseline characteristics for this cohort are shown in Table 1. The median age of AIS patients was 61 years (interquartile range [IQR] 57–65), where 73.0% were male and 97.3% were African American. Comorbid diagnoses upon admission included hypertension (74.3%), atrial fibrillation (13.5%), diabetes (13.5%), hyperlipidemia (24.3%), coronary artery disease (10.8%), and congestive heart failure (21.6%). A history of tobacco use was reported in 56.8% of patients. The median hemoglobin A1C was 5.8% (IQR 5.4–6.1), and the median low-density lipoprotein (LDL) cholesterol level was 95 mg/dL (IQR 75–117). Finally, the median NIHSS admission score was 7 (IQR 3–11), with a median baseline mRS score of 0 (IQR 0–2).

Per TOAST criteria, 39.2% of strokes were attributed to large-artery atherosclerosis, 20.3% to cardioembolism, 18.9% to small vessel disease, 17.6% to cryptogenic causes, and 4.1% to other determined causes. Furthermore, 10.8% of AIS patients received thrombolytics as part of treatment, and 18.9% underwent thrombectomy. The median NIHSS score at discharge was 3 (IQR 1–5), and the median mRS score at discharge or earliest follow-up was 2 (IQR 1–4). Overall mortality was quite low, with 2 patients (2.7%) expiring during their hospital admission.

Patient outcomes are stratified by TOAST classification in Table 2. Patients with small vessel disease had the highest median hemoglobin A1c of 5.9% (IQR 5.7–6.2), while those within the other determined cause category had the highest median LDL cholesterol of 107 mg/dL (IQR 92–113). Additionally, patients with strokes attributed to large vessel atherosclerosis had the high-

est median initial NIHSS scores (8 [IQR 6–10]) and highest median discharge NIHSS scores (3 [IQR 2–6]). Both of the patients who expired during admission were classified within the large artery atherosclerosis category.

In our study, 39.2% of patients met TOAST criteria classification for large artery atherosclerosis as the etiology for their strokes. Historically, cocaine-related strokes have been associated with reversible vasospasm in the small vasculature of the brain, as well

as platelet aggregation and increased strain on the cardiovascular system.² Yet, our findings raise the question of the significance of chronic inflammation from long-term cocaine use in precipitating ischemic strokes, as well as potential preventative measures for these patients.

Cocaine use has a potential impact on inflammation and the development of atherosclerosis through several pathways. Cocaine is known to increase levels of endothelin-1, a potent inflammatory protein that promotes atherosclerosis.⁶ It additionally increases expression of cell adhesion molecules like intracellular adhesion molecule-1 (ICAM-1), vascular cell adhesion molecule-1 (VCAM-1), and endothelial leukocyte adhesion molecule-1 (ELAM-1), which are known to facilitate leukocyte migration into blood vessels and contribute to plaque formation.⁷ Regular cocaine use is also known to cause the release of proteolytic substances from mast cells, which also accelerates atherosclerosis by degrading and facilitating uptake of LDL cholesterol by macrophages, leading to plaque formation.⁸ Figure 1 displays the known mechanisms of cocaine-mediated inflammation causing atherosclerosis in the brain.

It is also important to discuss safety and efficacy of thrombolytics and thrombectomy for this population. In our sample, eight patients received thrombolytics, and 14 underwent thrombectomy. Studies such as Siniscalchi et al.⁹ have examined the use of intravenous tissue plasminogen activator (tPA) in patients with documented cocaine-related AIS, with favorable benefits from administration of tPA and similar safety outcomes in comparison with their non-tPA counterparts. Furthermore, mechanical thrombectomy has widely been documented as an extremely important treatment in management of ischemic stroke patients, but there is very little research regarding the efficacy in cocaine-related AIS patients.¹⁰ Larger studies are needed to assess the effectiveness of both thrombolytics and thrombectomy within this unique patient population.

This is a single-center retrospective review, with no direct correlations able to be drawn from our results. However, our sample size of 74 AIS patients with documented cocaine use represents the largest cohort study to date for this patient population. Detailed patterns of cocaine consumption could not be analyzed because information was inconsistently recorded in the electronic medical record, likely reflecting both provider documentation practices and patient reluctance to disclose illicit drug use. Thus, this observational study provides insight into a specific cohort not frequently discussed in this literature and offers potential insight for future research endeavors.

In conclusion, this is the single largest cohort study in cocaine-related AIS conducted to date. Cocaine-related ischemic strokes may have a higher association with large artery atherosclerosis

Table 1. Baseline characteristics of AIS patients

Characteristic	AIS patients (n=74)
Age (yr)	61 [57–65]
Sex	
Male	54 (73.0)
Female	20 (27.0)
Race	
African American	72 (97.3)
White	2 (2.7)
Comorbid diagnoses	
Hypertension	55 (74.3)
Atrial fibrillation	10 (13.5)
Diabetes mellitus	10 (13.5)
Hyperlipidemia	18 (24.3)
coronary artery disease	8 (10.8)
Congestive heart failure	16 (21.6)
Substance use history	
History of tobacco use	42 (56.8)
Admitted history of cocaine use	17 (23.0)
Hemoglobin A1c (%)	5.8 [5.4–6.1]
LDL cholesterol (mg/dL)	95 [75–117]
Initial NIHSS	7 [3–11]
Baseline mRS	0 [0–2]
TOAST criteria classification	
Large artery atherosclerosis	29 (39.2)
Cardioembolic	15 (20.3)
Small vessel disease	14 (18.9)
Cryptogenic	13 (17.6)
Other determined cause	3 (4.1)
Treatment of stroke	
Thrombolytics	8 (10.8)
Thrombectomy	14 (18.9)
Discharge NIHSS	3 [1–5]
mRS at discharge or earliest follow-up*	2 [1–4]
Overall mortality	2 (2.7)

Values are presented as median [interquartile range] or n (%).

AIS, acute ischemic stroke; LDL, low-density lipoprotein; NIHSS, National Institutes of Health Stroke Scale; mRS, modified Rankin Scale; TOAST, Trial of ORG 10172 in Acute Stroke Treatment.

*mRS at discharge or earliest follow-up not recorded for 9 patients overall.

Table 2. Patient outcomes by TOAST criteria designation

	Large artery atherosclerosis (n=29)	Cardioembolic disease (n=15)	Small vessel disease (n=14)	Cryptogenic (n=13)	Other determined cause (n=3)
Hemoglobin A1c (%)	5.7 [5.3–6.2]	5.8 [5.5–6.5]	5.9 [5.7–6.2]	5.7 [5.4–6.0]	5.4 [5.2–5.8]
LDL cholesterol (mg/dL)	84 [75–110]	83 [70–97]	100.5 [83–134]	98 [93–130]	107 [92–113]
Initial NIHSS	8 [6–10]	7 [5–16]	3 [2–6]	4 [2–21]	0 [0–0]
Baseline mRS	0 [0–1]	2 [0–3]	0 [0–1]	0 [0–0]	0 [0–0]
Discharge NIHSS	3 [2–6]	3 [2–5]	2 [0–4]	3 [1–9]	0 [0–6]
mRS at discharge or earliest follow-up*	3 [1–4]	3 [1–4]	1 [1–3]	1 [1–3]	3 [1–4]
Overall mortality	2 (6.7)	0 (0)	0 (0)	0 (0)	0 (0)

Values are presented as median [interquartile range] or n (%).

TOAST, Trial of ORG 10172 in Acute Stroke Treatment; LDL, low-density lipoprotein; NIHSS, National Institutes of Health Stroke Scale; mRS, modified Rankin Scale.

*mRS at discharge or earliest follow-up was not recorded for 9 patients overall.

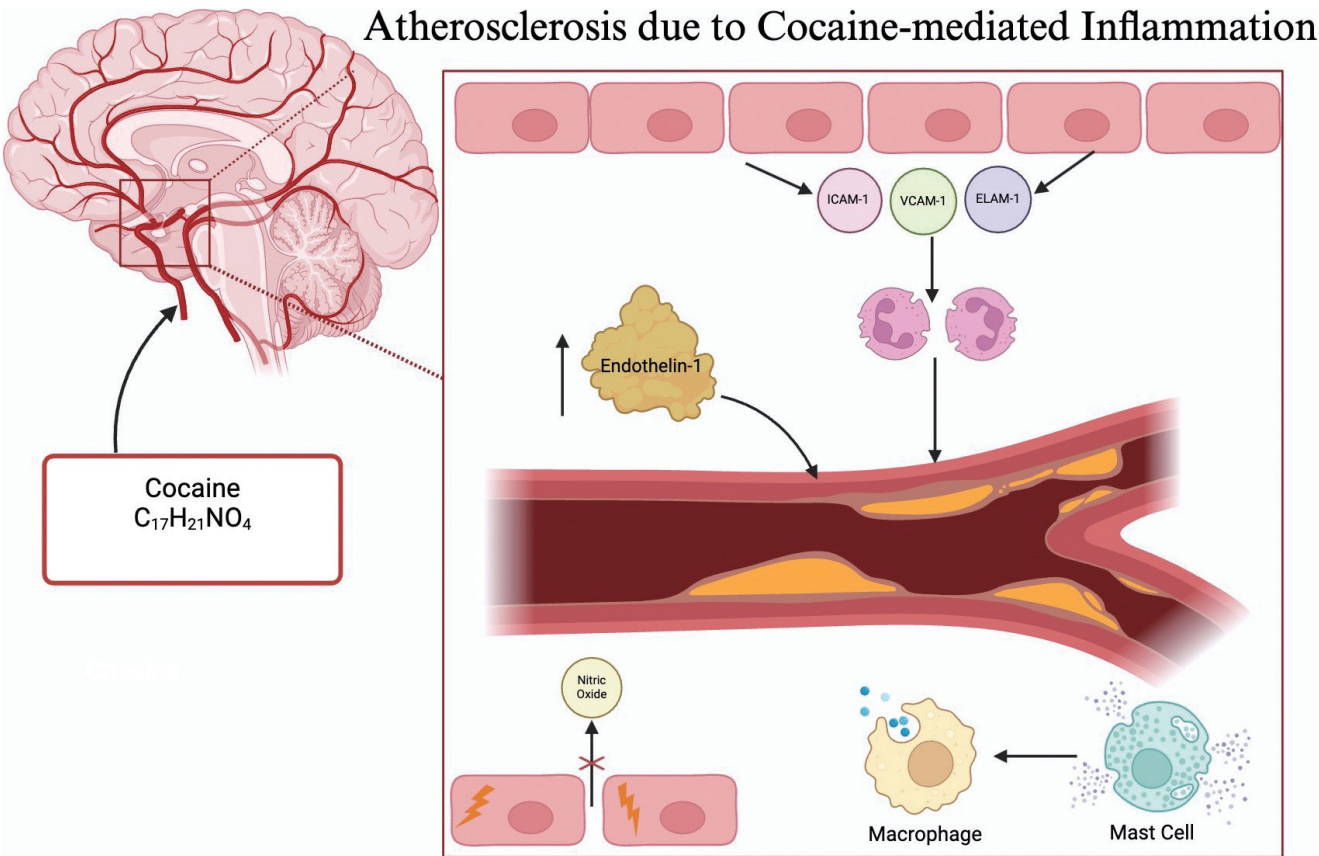


Figure 1. Atherosclerosis due to cocaine-mediated Inflammation. Cocaine use has a potential impact on inflammation and the development of atherosclerosis through several pathways. This figure demonstrates some of the known cocaine-mediated pathways that contribute to this inflammation. Created with BioRender.com, under BioRender's Academic License Terms.

than previously understood. Future studies are needed to further evaluate this potential mechanism, as well as contributing factors and treatment regimens such as thrombolytics and thrombectomy, within this unique population.

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Conflicts of interest

The authors have no financial conflicts of interest.

Author contribution

Conceptualization: SAK, TKH, JES, SJM. Study design: SAK, TKH, JES, SJM. Methodology: SAK, SJM. Data collection: SAK, SJM. Investigation: SAK, SJM. Statistical analysis: OAK, SAK. Writing—original draft: OAK, SAK. Writing—review & editing: all authors. Approval of final manuscript: all authors.

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